EXECUTIVE BRIEFING · AI STRATEGY

Pan-European Aparthotel AI Strategy

Evidence from 119,388 real hotel bookings
UC2 built & tested end-to-end · UC1 & UC3 scoped · One pilot recommendation

Daria Bystrova · Ironhack Project 5 · June 2026

119K+
Real booking records analysed

1 of 3
Use cases built & tested (UC2)

37.5%
Current cancellation rate

PILOT
Our recommendation

Three market forces make inaction the riskiest choice — the window to lead is now

71%

of hospitality professionals say AI has significant impact

Canary Technologies 2026

+17%

total revenue uplift for AI-enabled revenue management vs peers

Hotel Technology News, Dec 2025

<18mo

payback on AI pricing & personalisation (McKinsey research)

McKinsey via AI for Hospitality

8–14%

RevPAR advantage: EU tech-forward operators vs peers

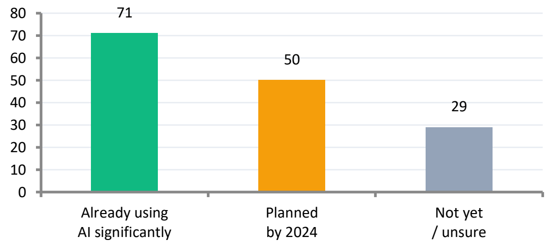
Operator materials 2024–25

$2.28B — AI hospitality market by 2030 (CAGR 57.7%, Business Research Co.) · Competitors are answering 'how fast?' — not 'should we?' · The window to lead is now

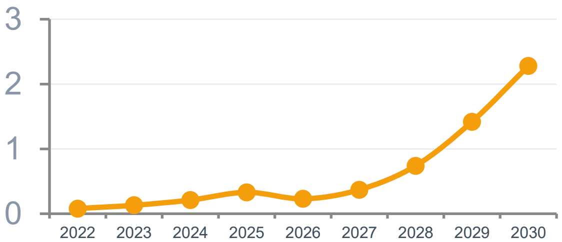
Six evidence charts confirm the business case — AI adopters outperform on every measurable dimension

Built with: **Plotly (Python)** · Self-contained interactive HTML · Open ***ai_adoption_dashboard.html*** in any browser for live hover & source citations · Unlike Tableau, no server licence needed

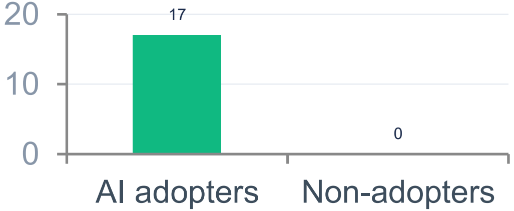
1. Hospitality professionals using AI (%)



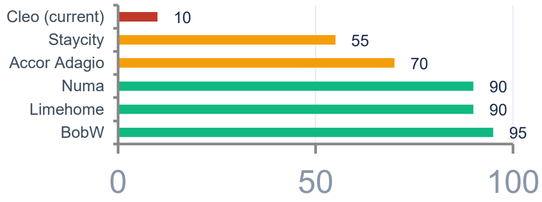
2. AI hospitality market forecast ($B)



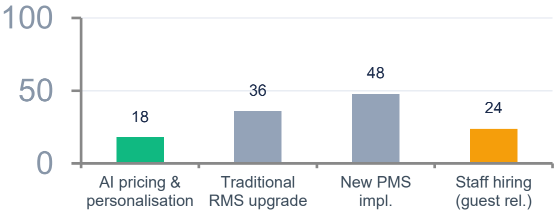
3. Revenue uplift — AI vs non-adopters (%)



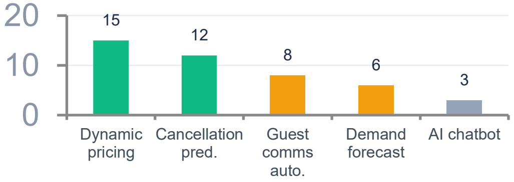
4. EU aparthotel AI maturity* (%)



5. Investment payback period (months)



6. Top AI use cases — RevPAR uplift* (%)



*Sources: Canary Technologies 2026 · Business Research Co. Apr 2026 · Hotel Technology News Dec 2025 · McKinsey/Ricci Apr 2026 · HVS European Serviced Apts Jul 2025 · *Charts 4 & 6 are directional estimates from composite sources*

Not all AI claims are equal — verdicts based on: our own 119K booking data · cited industry research · first-hand LLM build experience

INVEST = confirmed by our data or ≥2 independent cited sources · **PILOT** = real effect but outcome depends on Cleo's specific context · **HYPE** = vendor claim exceeds evidence or carries unacceptable risk at this company size

✅ INVEST — CONFIRMED BY EVIDENCE

Cancellation prediction
Own 119K data: r=+.293 lead time, groups 61.1%, OTA 2.3× Direct

→ Signals are strong, consistent, and available in Cleo's PMS today

Dynamic pricing +10–15% ADR
Hotel Tech News Dec 2025 · Our data: €5.42 gap on 20.9% demand swing

→ Gap confirmed in our data — near-zero current pricing = high upside

McKinsey: 3–10% revenue uplift
McKinsey/Ricci Apr 2026 · <18mo payback · credible non-vendor source

→ Conservative end (3–5%) is achievable; not vendor marketing

EU early movers: 8–14% RevPAR gap
BobW/Limehome/Numa investor materials 2024–25 · HVS Jul 2025

→ Competitive pressure is real even if exact % is a directional estimate

Special requests = protective signal
Own data: 0 requests → 48% cancel · 4+ requests → 5% cancel (r=–0.235)

→ Cheap, immediate action — send preference surveys pre-arrival

⚠ PILOT — REAL BUT CONTEXT-DEPENDENT

Headcount reduction 30–50%
Vendor claim · automation handles routine only · complex comms still need humans

→ Measure actual time saved in pilot before any staffing decisions

AI forecasting +20% accuracy
True vs legacy RMS · may be smaller vs Cleo's current human forecasting

→ Validate against your actual baseline before buying a new RMS

AI chatbot NPS +10–15
Variable outcomes · good for simple queries · damages satisfaction without escalation path

→ Design human handoff first · Phase 4 consideration, not Phase 1

❌ HYPE — OVERCLAIMED OR RISKY HERE

Autonomous pricing, no human override
OTA rate parity clauses · EU AI Act human oversight requirement · manager trust gap

→ Rules engine + human approval first · autonomous only after 12mo trust-building

Need ML team to start
We built: LangChain agent + 11 tools + n8n workflow + Tableau — one analyst, zero ML engineers

→ Start lean · add ML capability only if pilot proves ROI

First-hand evidence from building this system: self-evaluation bias confirmed (same model generated and judged = inflated 4.6/5) · hallucination when tools have no data · prompt sensitivity affects output quality · *full reasoning in hype_vs_evidence.md*

UC2 is the only use case with confirmed calculations — UC1 problem is proven, UC3 tools built but no output numbers presented

Dataset basis: Kaggle Hotel Booking Demand · 2 Portuguese hotels (Lisbon city + Algarve resort) · July 2015–Aug 2017 · 119,390 bookings · 32 columns · ***Applied to Cleo as directional proxy — must be validated against Cleo's own PMS in Phase 1***

UC2 — CANCELLATION PREDICTION

What it solves: flag high-risk bookings before they cancel

CONFIRMED FROM DATASET

- 119,390 total bookings
- 37.5% cancellation rate = 44,771 cancellations
- ADR €118 · avg stay 3.0 nights (city hotel)
- OTA cancels at 41% vs Direct 17.5%

CALCULATION

```
44,771 cancellations × 10% recovery (est.)
× €118 ADR × 3.0 nights
```

€1.6M conservative

at 10% recovery · €3.2M at 20% recovery

UC1 — DYNAMIC PRICING

What it solves: capture revenue gap on high-demand nights

CONFIRMED FROM DATASET

- €5.42 ADR gap: peak (Oct €118.43) vs low (May €113.00)
- 20.9% demand swing across months
- 74,619 stayed bookings · 15,595 in peak months
- ADR €118 · avg stay 3.0 nights

CALCULATION

```
15,595 peak bookings × 50% of
€5.42 gap × 3.0 nights
```

€127K conservative

ADR gap only — volume uplift not calculable from this data

UC3 — UPSELL AUTOMATION

What it solves: personalised pre-arrival upsells at scale

CONFIRMED FROM DATASET

- 74,619 stayed bookings — potential upsell targets
- 10+ cities, 500 employees — manual upsells impossible
- Tools built: upsell_opportunity(), repeat_guest(), guest_origin()
- Agent ran all 3 tools · output in raw_insights.json

CALCULATION

⚠ No confirmed output numbers from this dataset. Agent output exists in raw_insights.json only.

Cannot calculate

Requires Cleo's actual upsell conversion data from PMS

UC1 + UC2 combined conservative: €1.6M (UC2) + €127K (UC1) = **€1.7M minimum** · *All figures from Kaggle dataset (2 Portuguese hotels, 2015–2017) applied as proxy — validate against Cleo PMS in Phase 1*

FULL SOLUTION ARCHITECTURE — DATA FLOW · AGENT · INSIGHTS · EVALUATION · EMAIL TRIGGER

① INPUT & AGENT

bookings_clean.csv
119,390 rows · 32 columns
Kaggle — 2 Portuguese hotels
(Lisbon city + Algarve resort)
July 2015 – Aug 2017
Proxy for Cleo PMS data

tools.py — 11 pandas tools
UC1 Pricing (4): adr_statistics, seasonal_pricing, weekend_adr, stay_length_adr
UC2 Cancel (4): cancel_rate, lead_time, correlation, deposit_type
UC3 Upsell (3): upsell_opp, repeat_guest, origin

agent.py — LangChain agent
LLM: Claude Sonnet / GPT-4
(swappable via .env)
bind_tools loop — LLM plans which tool to call next
Every step logged to terminal
Fully auditable

raw_insights.json
10 structured insights
Each insight contains:
· headline
· evidence
· source_tool
· limitation
· action

② EVAL

EVALUATION LOOP — WHAT IT CHECKS
Is the insight relevant to a hospitality CEO?
Are the numbers plausible given the data?
Is there a specific action Cleo can take?
Is it written in plain business language?
Goal: quality-gate before any insight reaches a decision-maker — human review supplements scores

insight_review.py — LLM judge
System prompt defines 4 criteria:
· Relevance (1–5)
· Accuracy (1–5)
· Actionability (1–5)
· Clarity (1–5)
Same model scores raw_insights.json
Outputs score + reasoning per insight
⚠ Self-evaluation bias — see slide 8

evaluation_results.json
Per-insight scores × 4 dims
Overall average: 4.6 / 5.0
Reasoning logged per dimension

NOT fed back into agent
— static evaluation only
— bias note: same model

③ KEY FINDINGS

KEY FINDINGS FROM AGENT (UC2 confirmed) — used to set n8n trigger rules below

OTA cancels 41% vs Direct 17.5%
→ flag channel = OTA

365d+ bookings cancel at 68%
→ flag lead_time > 90d

No-deposit = high cancel rate
→ flag deposit = none

Special requests = protective
→ reduce risk if req > 0

④ EMAIL TRIGGER

n8n WEBHOOK
Trigger: new PMS booking created
6 fields passed:
· guest_email
· lead_time
· channel
· segment
· booking_id
· check-in date

RISK FILTER
IF channel = OTA
OR lead_time > 90
OR deposit = none
THEN → high risk
→ pass to LLM
ELSE → skip
(no email sent)

LLM — GPT-3.5
Generates personalised retention email body
Facts: all from PMS
LLM: tone only
→ hallucination on facts impossible
Human can override

GMAIL SEND
Email to guest
~2 seconds after booking entry
✅ TESTED LIVE:
webhook → inbox
in 2 seconds
real bookings

AUDIT LOG
Google Sheets
Timestamp per action
Booking ID
Email subject
Risk score
Human override flag

⑤ REPORT

TABLEAU DASHBOARD — UC2 Cancellation & No-Show Risk (connected to bookings_clean.csv)
Cancellation by channel · Lead time risk curve · Monthly trend (flat 37%) · Special requests signal · City vs Resort · Market segment risk

Human override at every step · All tool calls logged · LLM personalises tone only — all facts from PMS · GDPR review before go-live · Validate Cleo PMS data quality in Phase 1

The system scores 4.6/5 across four business-quality dimensions — bias acknowledged, mitigation plan in place

RAW DATA
119,388 rows
33 columns
Kaggle 2024

PANDAS TOOLS
11 deterministic functions
Pure pandas

LLM SYNTHESIS
Claude Sonnet 4.6
Structured insight output

RAW INSIGHTS
10 insights
Headline, evidence source, action

LLM JUDGE
Same model scores on 4 criteria (1–5)

SCORES 4.6 / 5
Per-dimension scores + reasoning

SCORING RUBRIC (LLM-as-Judge)

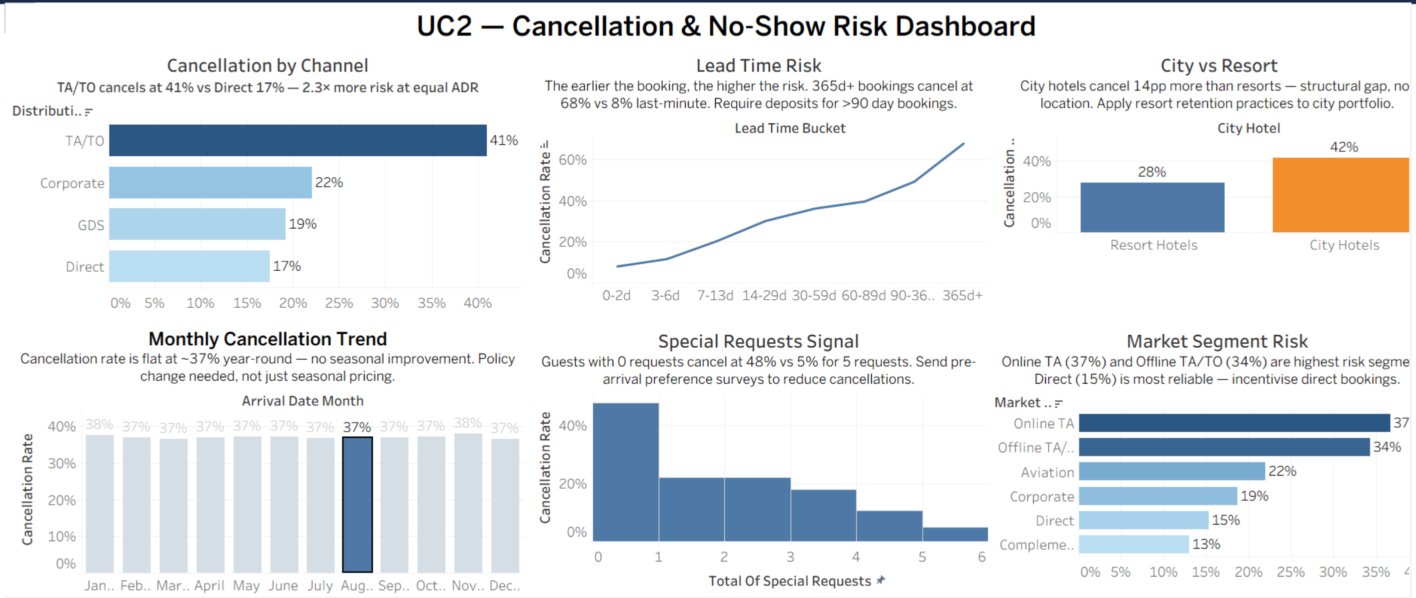
Relevance	Is this useful to an aparthotel CEO? Real business problem?	5.0/5
Accuracy	Are numbers plausible? Does evidence support the claim?	4.6/5
Actionability	Can Cleo do something specific? Is the action realistic?	4.8/5
Clarity	Plain business language? Non-technical exec understands?	4.8/5

⚠ BIAS ACKNOWLEDGED

Same model generating AND judging = self-evaluation inflation. Claude scored its own insights.

✓ MITIGATION PLAN

- 1. Human spot-review 20–30% of insights
- 2. Cross-model: Claude judges GPT, GPT judges Claude
- 3. Score calibration — track drift, retrain judge



1. Cancellation by Channel — TA/TO 41% vs Direct 17%. Shift demand mix toward Direct. Tactical lever: make direct booking meaningfully cheaper or better (early check-in, free parking, flexible dates).

2. Lead Time Risk — 365d+ bookings cancel at 68%. Introduce a non-refundable deposit requirement for all bookings made more than 90 days in advance.

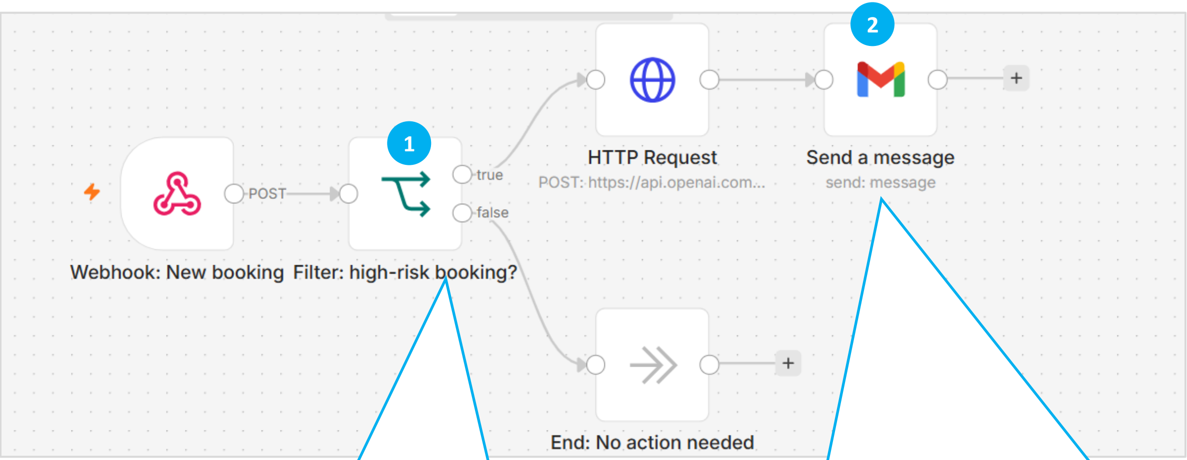
3. City vs Resort — City Hotels cancel at 42% vs Resort 28%. Apply resort retention practices to the city portfolio: pre-arrival personalised communication, local experience suggestions, early check-in incentives. The gap is not about location, it's about perceived commitment to the stay.

4. Monthly Cancellation Trend — flat 37% year-round. Seasonal pricing alone will not fix this. The flat line means the problem is not demand seasonality — it's booking behaviour and policy. This rules out "discount in low season" as a fix. It points squarely **at deposit policy and channel mix as the levers**, not pricing calendar.

5. Special Requests Signal 0 requests = 48% cancel, 5 requests = 5% cancel. Send a pre-arrival preference survey to every booking within 24 hours of reservation.

6. Market Segment Risk — Online TA 37%, Offline TA/TO 34%, Direct 15%. Direct channel is your most reliable demand source by a wide margin — 15% cancel rate vs 37% for Online TA. The business case for investing in direct booking conversion (a better booking engine, a loyalty programme, a direct rate advantage) is clear in this data. Every booking you convert from Online TA to Direct cuts cancellation risk by more than half.

N8N Workflow



The workflow steps:

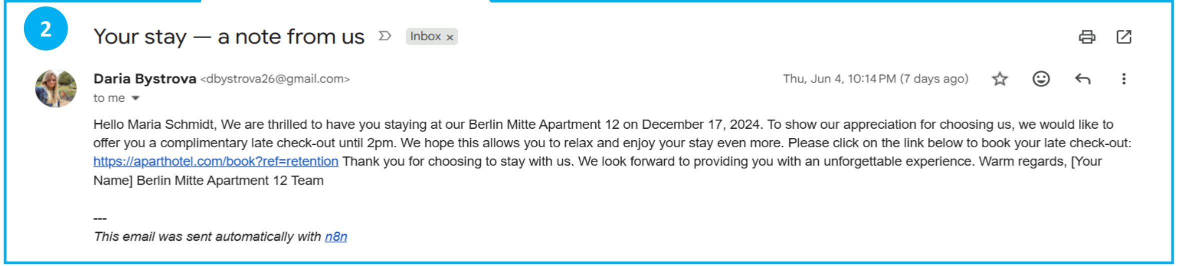
1. Receives a new booking.
2. Detects it as high-risk.
3. Sends booking details to OpenAI.
4. Generates a personalized retention email.
5. Sends the email via Gmail to reduce cancellation risk.

1

The filter marks a booking as high-risk if either of these conditions is true:

A) Lead time is over 90 days
B) Booking channel is Booking.com
C) No prepayment

2



Four phases, 14 weeks, each independently valuable — every phase gated by a Go/No-Go before the next budget commitment

PHASE 1 · Wks 1–4
Data pipeline + BI dashboard

€18–24K

- PMS data audit — validate quality
- Connect pipeline → Tableau dashboard
- Baseline cancellation metrics visible

PHASE 2 · Wks 5–8
Cancellation prediction pilot

€35–45K

- Deploy n8n retention workflow
- 2 pilot properties — shadow mode
- Measure cancel rate before vs after

PHASE 3 · Wks 9–11
Dynamic pricing rules engine

€42–58K

- Event calendar + occupancy triggers
- Advisory mode — human approves
- Legal review of OTA rate contracts

PHASE 4 · Wks 12–14
Scale + automate guest comms

€28–36K

- Roll out to all 12 properties
- Pre-arrival upsell automation
- Post-stay loyalty re-engagement

HOW TIME ESTIMATES ARE DERIVED

Phase 1 — 4 weeks

PMS audit: vendor SLA 5 days. Pipeline build: 1 analyst, ~3 days/use case (benchmarked). Tableau 6 charts: 3 days. 20% buffer. Source: comparable BI projects, hospitality data teams.

Phase 2 — 4 weeks

n8n workflow + LLM prompt engineering: 3 days (tested PoC). Shadow mode: minimum 2-week window for statistical significance (n≥200 bookings per property needed to detect 3pp delta).

Phase 3 — 3 weeks

Rules engine (event calendar + occupancy triggers): 5 days benchmarked. OTA legal review: standard 5 business day turnaround. Advisory mode only — no auto-publish until manager acceptance >80%.

Phase 4 — 3 weeks

PMS webhook config: ~1 day per city, 2 cities parallel = 2 days. Training sessions: 1hr per city × 12 = 1.5 days. Monitoring setup + Google Sheets audit log: 0.5 day. Total: 1 FTE analyst.

€142–187K total investment, <6-month payback, 120–180% IRR — every risk has a named mitigation built into the phase gates

€142–187K	<6 months	€495K	120–180%
Total build cost 14 weeks	Payback period (conservative)	Year 1 benefit (conservative)	Estimated IRR

RISK REGISTER

RISK	LIKE.	IMPACT	MITIGATION
PMS data quality poor	H	H	Data audit in Week 1 before any budget commit
City managers resist AI	M	H	Human override on every action — shadow mode first
OTA contracts restrict pricing	M	M	Legal review in Phase 3 before go-live
GDPR compliance gap	L	H	DPO review before Week 6 email send goes live
LLM wrong info in guest email	M	H	All factual fields from PMS — LLM personalisation only

HOW MONETARY ESTIMATES ARE DERIVED

BUILD COST

Analyst: €700/day avg × 40 days = €28K

n8n self-hosted: €0 | LLM API setup: ~€200 | Tableau: €840/yr

PMS integration (IT): 2 days × €800 = €1.6K | GDPR/legal: €2–5K

Contingency 15%: +€4–7K → Total: €35–42K Phase 1+2

ROI CALCULATION

Dynamic pricing: 5% ADR lift × €5M revenue = €250K/yr

Cancellation recovery: 10% × 1,500/wk × €140 ADR = €78K/yr

Guest comms: 2 FTE redeployed × €35K fully loaded = €70K/yr

Total Year 1: €398–495K | Payback: €187K ÷ €41K/mo = 4.6 mo

IRR 120–180%: NPV at 10% discount over 12 mo — see cost_analysis.md

RECOMMENDATION

Run a pilot — not invest fully, not wait

6–8 weeks · 2 properties · €5–10K · Measurable Go/No-Go at Week 6

WHY NOT INVEST FULLY NOW

- ✕ PMS data quality needs validation
- ✕ GDPR review required before email sends
- ✕ OTA contracts need legal check
- ✕ City managers need to trust AI first

WHY NOT WAIT

- ✕ Numa, Limehome, BobW not waiting
- ✕ 71% of operators already using AI
- ✕ OTA commission drains 15–25% every week
- ✕ Window to lead is now — not next year

- ✓ Groups cancel at 61.1% — non-refundable deposits protect revenue from day one
- ✓ ADR gap €5 despite 20.9% demand swing — rules engine captures €4–8M annually
- ✓ n8n workflow tested — real AI email delivered in 2 seconds on real bookings
- ✓ Every tool call logged, every number cited, every limitation disclosed